

V3 empirical study · Instruction

Seven models · four underlyings · four volatility regimes · 1.5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of seven models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t . That row is bold and shaded green; Ranking ranks 1–7.
- Fair model ranking is split: return-based models in the *_return_based stock PDF, Heston / Heston-Merton in the *_option_implied stock PDF.

Experimental design (held fixed for every model)

- Models: GBM, MD-GBM (Markov Directional Geometric Brownian Motion), GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- Companies: SPY, AAPL, MSFT, AMZN. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- P-measure: keep the lookback drift μ . Do not replace $\mu \rightarrow r$ (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

Seven models · four underlyings · four volatility regimes · 1.5-year monthly calibration

Path construction

- One rolling path cloud per ticker $\times$ regime $\times$ model, $n_paths = 10000$, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0 . Refresh parameters at month-ends with data $\leq$ that date.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: $RMSE\% = 100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- $MAE\% = 100 \times \text{mean}(|p50 - S_t|/S_t)$. $Bias\% = 100 \times \text{mean}((p50 - S_t)/S_t)$.
- Tables report RMSE%, MAE%, Bias%, and Ranking only.

Page 3 · Table block · SPY · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	25.80	20.35	+19.23	1
MD-GBM	27.08	21.76	+21.12	4
GARCH	33.58	28.91	+28.54	7
Heston	26.46	20.99	+20.03	3
Merton	27.61	22.46	+21.84	5
GARCH-Merton	33.42	28.77	+28.34	6
Heston-Merton	25.88	20.28	+19.02	2

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	4.49	3.72	+3.72	3
MD-GBM	4.67	3.91	+3.90	4
GARCH	8.02	6.91	+6.91	7
Heston	4.48	3.72	+3.71	2
Merton	7.02	5.94	+5.94	6
GARCH-Merton	5.37	4.56	+4.55	5
Heston-Merton	4.25	3.52	+3.52	1

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	8.90	7.69	+7.69	1
MD-GBM	9.21	8.05	+8.05	2
GARCH	24.77	22.96	+22.96	7
Heston	9.68	8.49	+8.49	4
Merton	13.33	12.63	+12.63	6
GARCH-Merton	12.22	11.55	+11.55	5
Heston-Merton	9.58	8.29	+8.29	3

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	8.57	6.67	-1.91	3
MD-GBM	8.40	6.36	-1.25	1
GARCH	20.26	14.62	+14.02	7
Heston	8.41	6.40	-1.44	2
Merton	9.02	6.30	+0.60	5
GARCH-Merton	12.93	10.37	+5.68	6
Heston-Merton	8.61	6.67	-1.93	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 4 · Table block · Company summary · seven models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green = lowest mean RMSE% vs realized S_t. 1.5-year lookahead, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis

2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	36.04	28.00	+19.79	3
MD-GBM	39.57	30.27	+25.99	5
GARCH	51.63	44.05	+41.08	6
Heston	36.24	28.22	+19.90	4
Merton	35.73	28.75	+16.53	2
GARCH-Merton	56.38	48.80	+46.41	7
Heston-Merton	35.08	27.64	+17.62	1

Table Co-2 Company summary · 2013-2014 Normal

2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	19.27	17.15	+5.06	5
MD-GBM	19.41	17.31	+7.00	6
GARCH	17.99	16.31	+7.93	1
Heston	19.03	16.95	+5.60	3
Merton	18.87	16.73	+10.39	2
GARCH-Merton	20.87	18.68	+6.41	7
Heston-Merton	19.04	16.93	+4.80	4

Table Co-3 Company summary · 2018-2019 Late-cycle

2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	26.51	24.33	+24.32	4
MD-GBM	28.46	26.36	+26.35	5
GARCH	50.42	46.52	+46.51	7
Heston	26.10	23.89	+23.89	3
Merton	25.40	23.03	+23.02	2
GARCH-Merton	40.72	37.55	+37.54	6
Heston-Merton	25.26	22.91	+22.90	1

Table Co-4 Company summary · 2019-2020 COVID

2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	19.09	15.67	-14.78	6
MD-GBM	17.66	14.50	-13.26	3
GARCH	9.81	7.89	-0.47	1
Heston	18.80	15.40	-14.45	5
Merton	18.69	15.22	-14.20	4
GARCH-Merton	11.20	9.40	-3.85	2
Heston-Merton	19.53	15.99	-15.11	7

RMSE%, MAE%, and Bias% are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-simulation of a combined price series. Ranking uses the mean RMSE% only.

Page 5 · Table block · AAPL · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	51.72	40.81	+37.11	1
MD-GBM	57.33	46.16	+44.38	5
GARCH	88.07	79.08	+78.06	6
Heston	53.57	42.24	+39.16	3
Merton	56.69	44.80	+42.41	4
GARCH-Merton	99.38	89.70	+88.82	7
Heston-Merton	51.96	41.02	+36.75	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	18.03	15.26	-14.06	6
MD-GBM	16.29	13.94	-12.31	4
GARCH	13.30	11.75	-9.22	2
Heston	17.48	14.90	-13.26	5
Merton	11.21	10.07	-6.75	1
GARCH-Merton	14.97	12.99	-10.92	3
Heston-Merton	18.37	15.64	-13.93	7

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	30.71	27.14	+27.12	4
MD-GBM	32.15	28.78	+28.76	5
GARCH	46.80	43.80	+43.78	7
Heston	30.62	27.11	+27.09	3
Merton	29.96	26.36	+26.34	2
GARCH-Merton	40.39	37.68	+37.66	6
Heston-Merton	29.66	25.87	+25.84	1

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	26.10	23.43	-23.41	6
MD-GBM	24.58	22.00	-21.96	4
GARCH	10.87	9.11	-3.36	1
Heston	25.43	22.72	-22.70	5
Merton	24.55	21.92	-21.87	3
GARCH-Merton	11.40	9.42	-7.05	2
Heston-Merton	26.13	23.30	-23.28	7

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 6 · Table block · MSFT · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	18.26	15.35	+6.51	4
MD-GBM	19.39	16.24	+9.55	6
GARCH	18.01	15.09	+8.32	3
Heston	17.91	15.09	+6.02	2
Merton	18.66	15.83	+4.33	5
GARCH-Merton	20.00	16.39	+11.42	7
Heston-Merton	17.83	15.04	+5.10	1

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	4.49	3.85	-3.08	5
MD-GBM	3.46	2.88	-1.80	3
GARCH	3.23	2.65	-1.51	2
Heston	4.06	3.42	-2.44	4
Merton	2.73	2.11	-0.04	1
GARCH-Merton	8.09	6.81	-6.08	7
Heston-Merton	4.53	3.84	-2.95	6

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	12.44	10.84	+10.84	4
MD-GBM	13.52	12.21	+12.21	5
GARCH	25.64	24.40	+24.40	7
Heston	12.11	10.39	+10.39	2
Merton	12.29	10.16	+10.16	3
GARCH-Merton	17.28	16.42	+16.42	6
Heston-Merton	11.85	9.90	+9.90	1

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	10.72	8.82	-7.67	5
MD-GBM	9.26	7.59	-6.14	3
GARCH	6.27	4.05	+1.88	1
Heston	10.50	8.63	-7.46	4
Merton	10.88	9.06	-8.18	6
GARCH-Merton	8.38	7.02	-5.84	2
Heston-Merton	11.33	9.28	-8.20	7

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 7 · Table block · AMZN · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	38.14	27.84	+15.74	4
MD-GBM	41.99	28.41	+24.03	5
GARCH	48.79	38.00	+36.87	6
Heston	37.25	27.34	+14.53	3
Merton	31.83	25.62	+2.85	1
GARCH-Merton	49.75	40.30	+38.98	7
Heston-Merton	35.47	26.87	+11.00	2

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	35.30	32.34	+32.32	2
MD-GBM	38.47	35.12	+35.11	5
GARCH	37.43	34.53	+34.53	4
Heston	35.55	32.51	+32.50	3
Merton	42.68	38.00	+37.97	7
GARCH-Merton	39.54	36.24	+36.22	6
Heston-Merton	34.22	31.32	+31.29	1

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	36.37	34.99	+34.99	4
MD-GBM	39.72	38.08	+38.08	5
GARCH	78.80	71.35	+71.35	7
Heston	35.57	34.18	+34.18	3
Merton	33.94	32.57	+32.57	1
GARCH-Merton	64.50	58.55	+58.55	6
Heston-Merton	34.27	32.96	+32.96	2

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
GBM	20.44	14.77	-13.27	4
MD-GBM	19.14	13.91	-11.69	3
GARCH	12.30	10.52	+0.06	1
Heston	20.48	14.84	-13.20	5
Merton	20.65	14.66	-12.56	6
GARCH-Merton	13.83	11.75	+1.36	2
Heston-Merton	21.14	15.38	-13.86	7

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Heston-Merton	21.57	19.75	5
2	Heston	21.85	19.19	0
3	GBM	21.90	19.35	3
4	Merton	22.06	19.65	4
5	MD-GBM	22.79	19.26	1
6	GARCH-Merton	28.21	16.12	0
7	GARCH	29.76	22.52	3

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	2	1	0	0	3
MD-GBM	1	0	0	0	1
GARCH	0	1	1	1	3
Heston	0	0	0	0	0
Merton	0	1	1	2	4
GARCH-Merton	0	0	0	0	0
Heston-Merton	1	1	2	1	5

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	33.48	15.58	22.11	16.46	21.90
MD-GBM	36.45	15.72	23.65	15.34	22.79
GARCH	47.11	15.49	44.00	12.43	29.76
Heston	33.80	15.39	22.00	16.21	21.85
Merton	33.70	15.91	22.38	16.28	22.06
GARCH-Merton	50.64	16.99	33.59	11.63	28.21
Heston-Merton	32.78	15.34	21.34	16.80	21.57

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	11.94	8.73	2
MD-GBM	12.34	8.81	1
GARCH	21.66	22.52	0
Heston	12.26	9.05	0
Merton	14.24	11.17	0
GARCH-Merton	15.98	12.57	0
Heston-Merton	12.08	9.09	1

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	31.64	28.41	1
MD-GBM	32.59	28.36	0
GARCH	39.76	30.05	1
Heston	31.78	28.03	0
Merton	30.60	27.26	1
GARCH-Merton	41.53	27.68	0
Heston-Merton	31.53	27.89	1

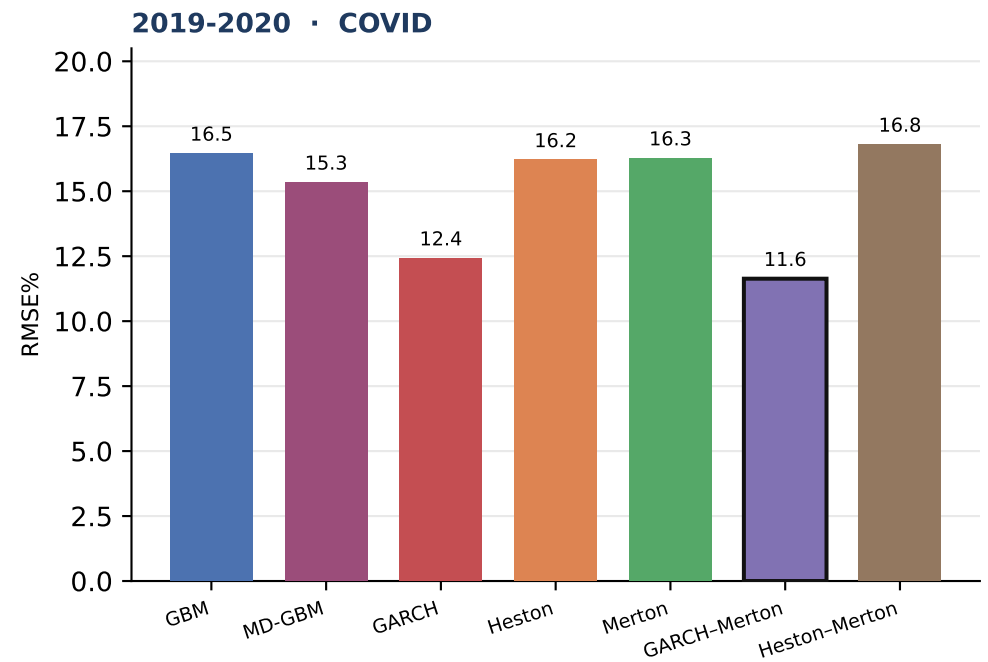
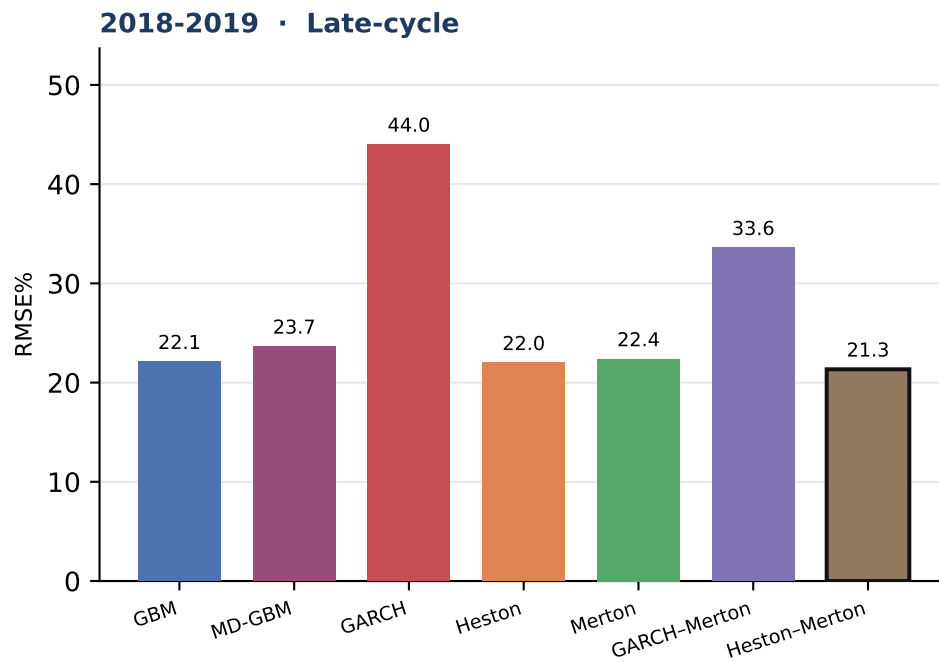
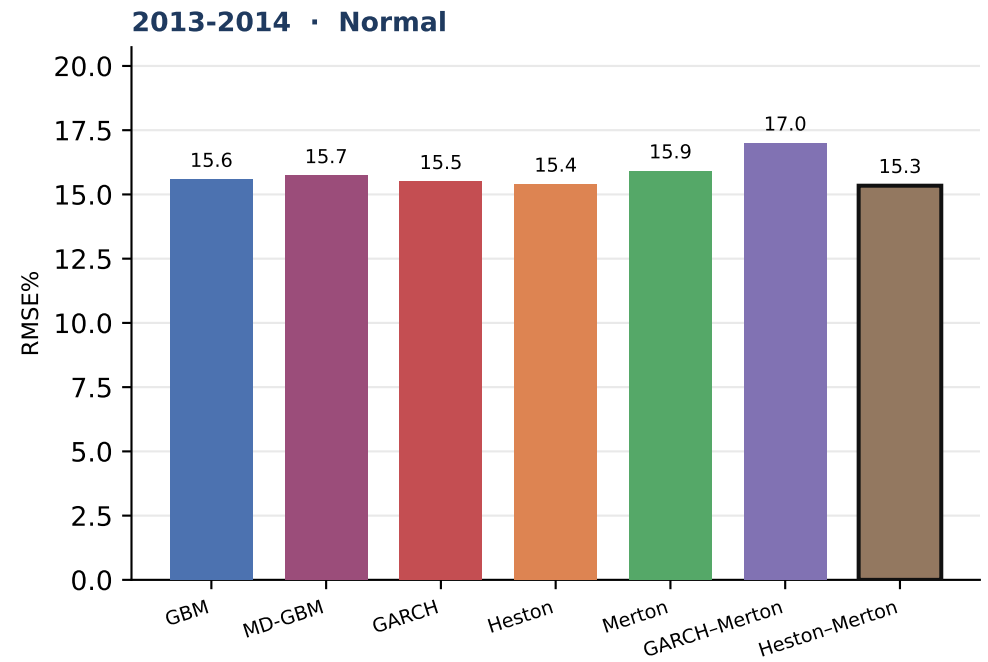
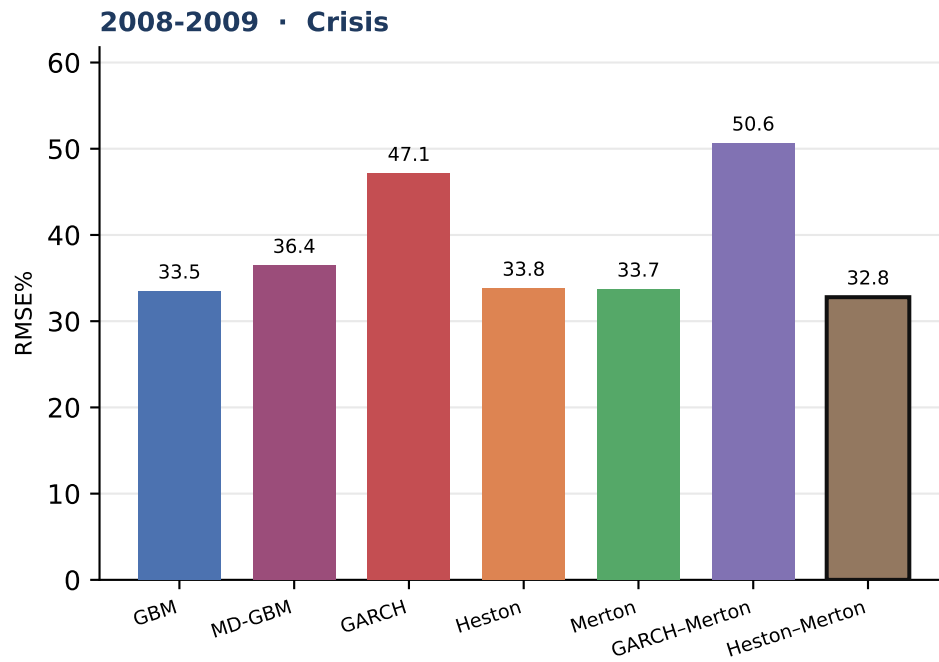
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	11.48	11.58	0
MD-GBM	11.41	11.39	0
GARCH	13.29	12.14	1
Heston	11.14	11.31	0
Merton	11.14	11.58	1
GARCH-Merton	13.44	12.83	0
Heston-Merton	11.39	11.59	2

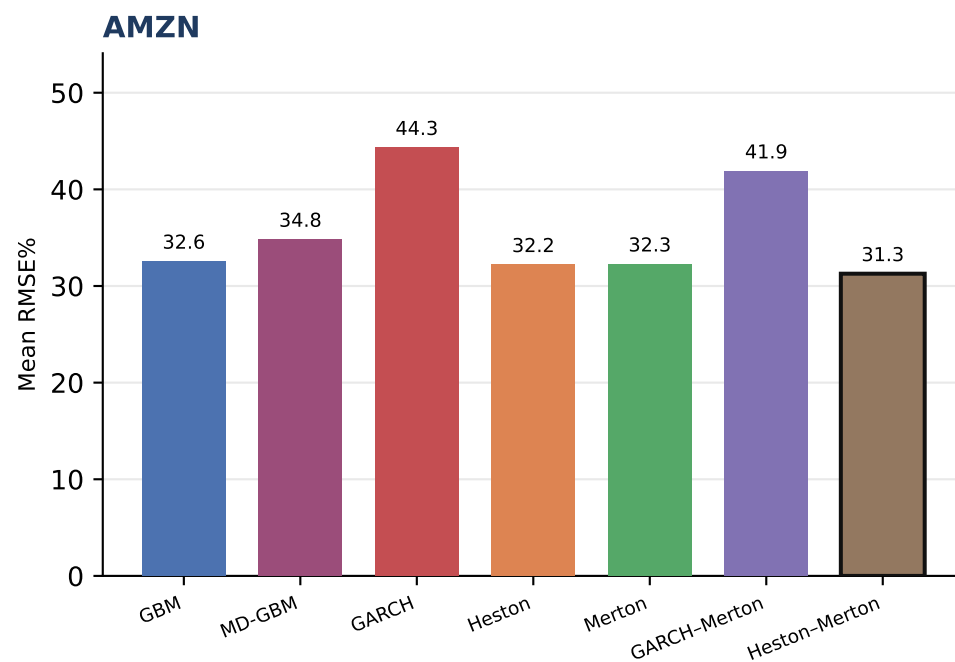
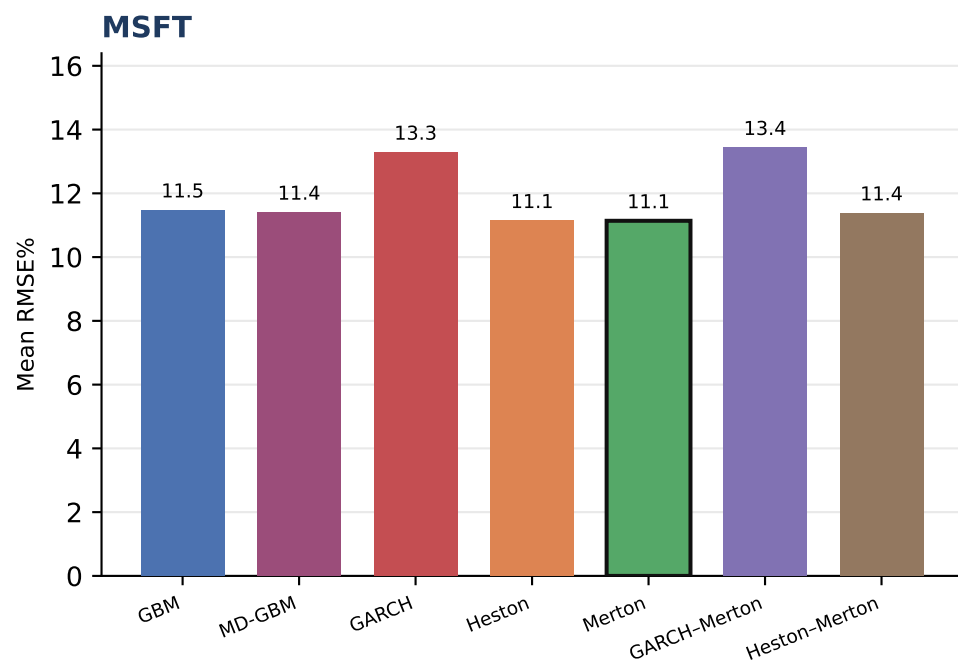
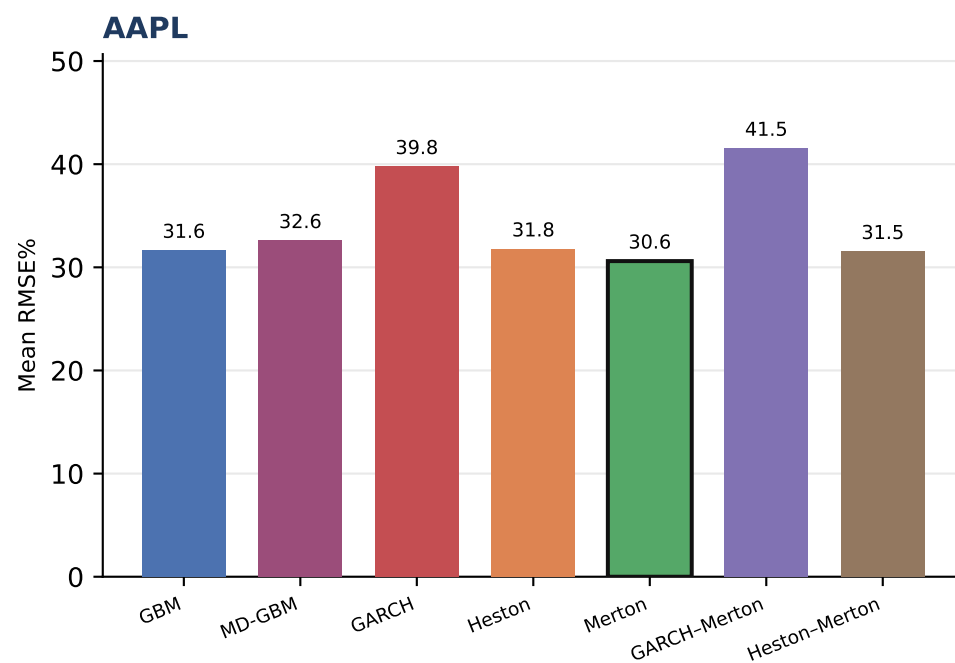
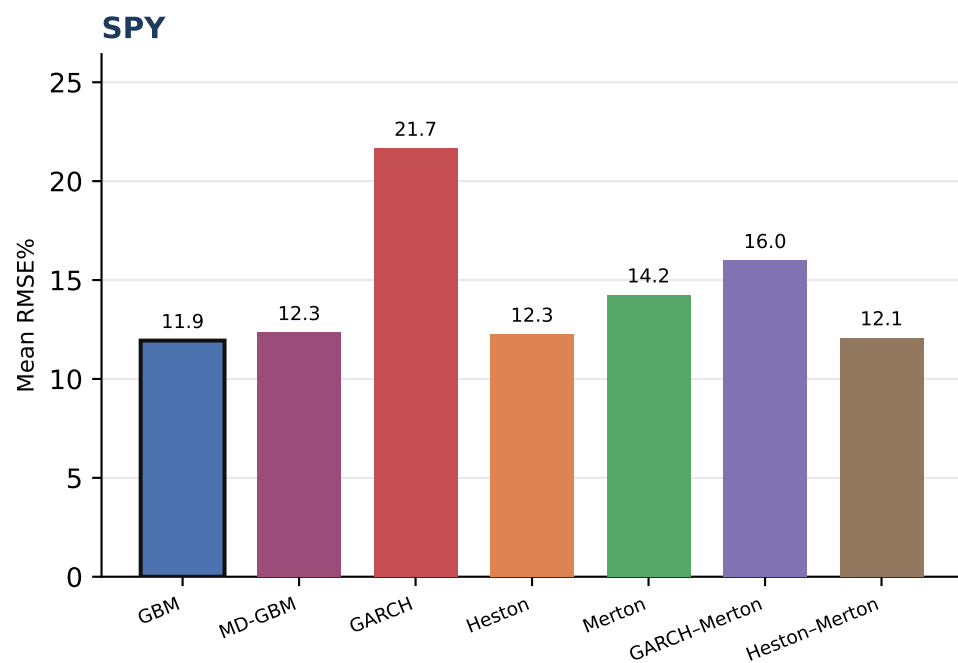
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	32.56	35.83	0
MD-GBM	34.83	39.10	0
GARCH	44.33	43.11	1
Heston	32.21	35.56	0
Merton	32.27	32.88	2
GARCH-Merton	41.90	44.64	0
Heston-Merton	31.27	34.25	1

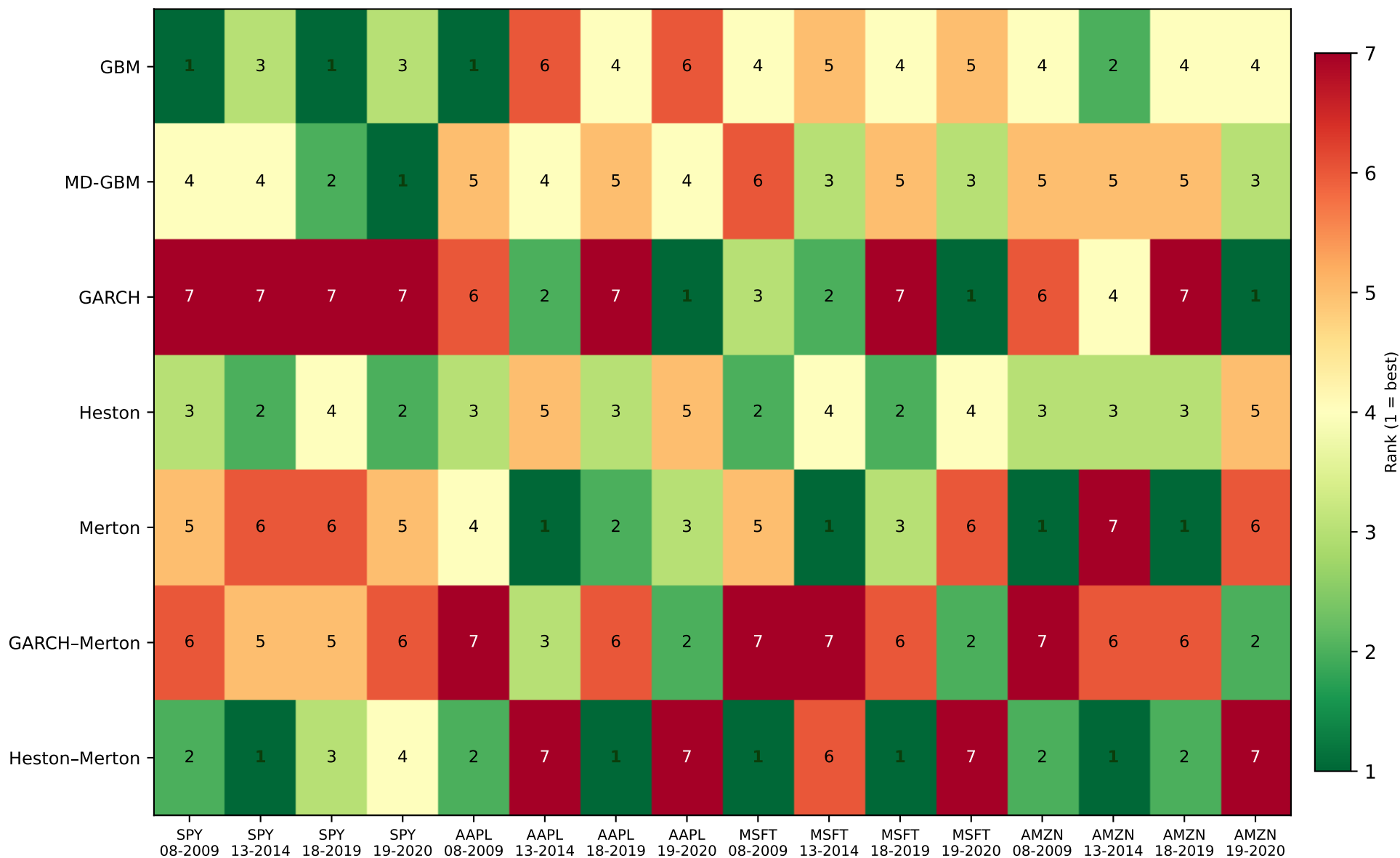
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



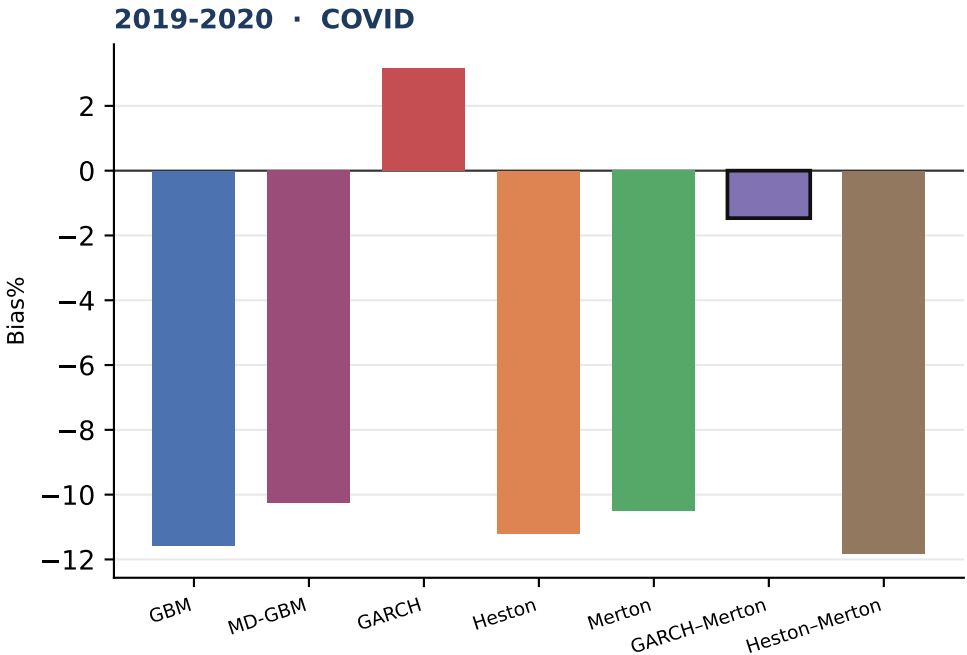
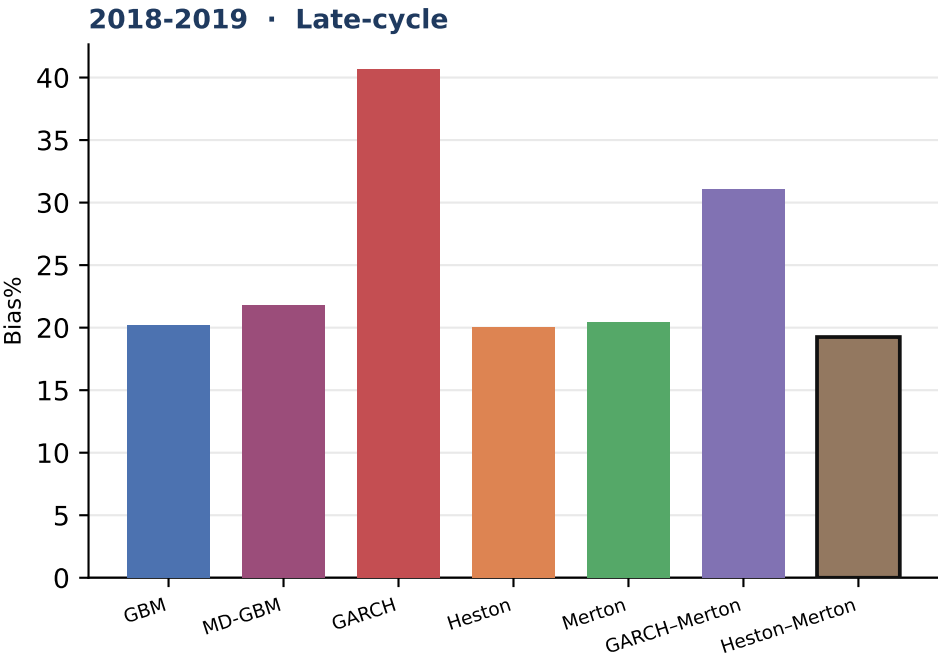
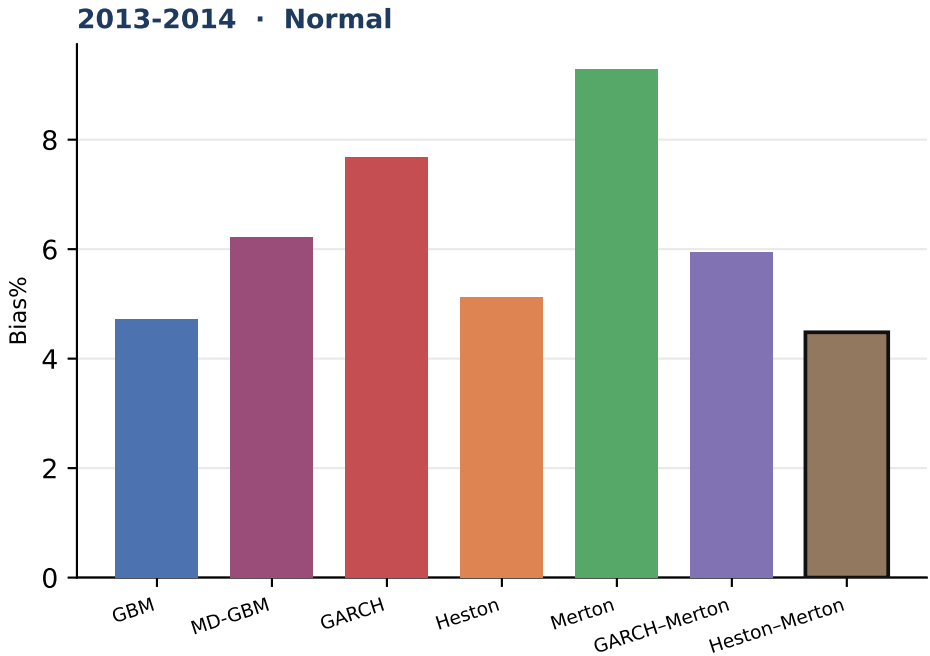
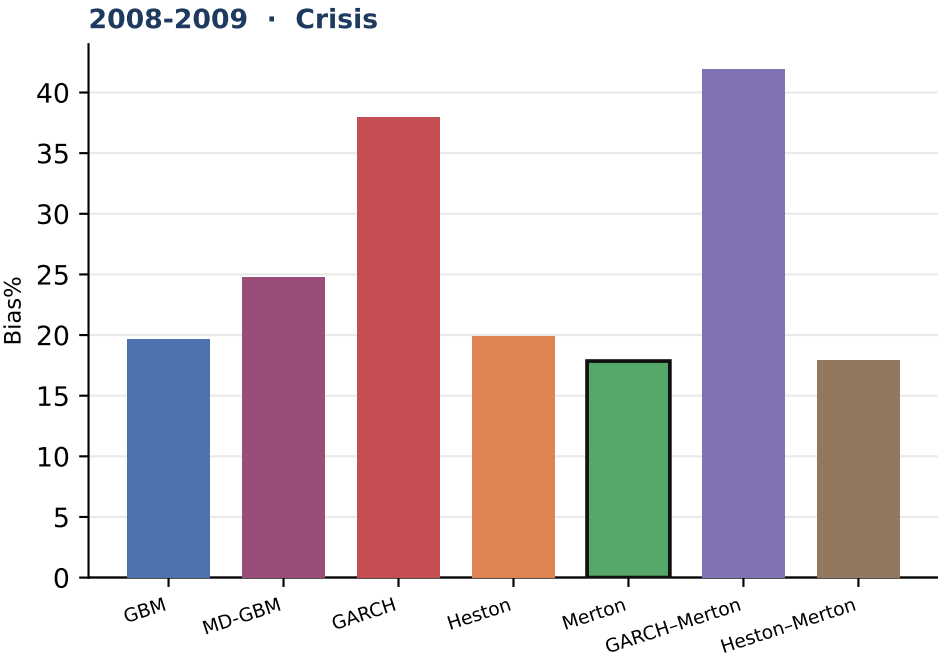
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero Bias% (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 16 company×regime cells, Heston–Merton is best (mean RMSE% = 21.57; 5 of 16 cells).
- The ranking is not stable. Best-in-cell models: GARCH, GBM, Heston–Merton, MD-GBM, Merton.
- These conclusions are conditional on 1.5-year monthly calibration P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.